

Core Execution & Workflow Patterns

- **Pattern 1: Prompt Chaining** [[00:54](#)] - Breaking large tasks into sequential, smaller steps where each step validates the previous output, much like an assembly line.
- **Pattern 2: Routing** [[05:42](#)] - Acting as a smart operator that analyzes user intent and directs the request to the most appropriate, specialized agent.
- **Pattern 3: Parallelization** [[09:30](#)] - Splitting a large goal into independent sub-tasks, executing them simultaneously across multiple agents, and then merging the normalized results.

Quality & Tooling

- **Pattern 4: Reflection** [[13:16](#)] - Using a critic agent to review a first draft against quality rubrics, generating feedback to revise and improve the output repeatedly.
- **Pattern 5: Tool Use** [[15:51](#)] - Enabling the AI to discover, authorize, and execute external tools (like APIs or web browsers) to retrieve real-world data.
- **Pattern 13: Retrieval (RAG)** [[36:01](#)] - Parsing, chunking, and embedding documents to create searchable vector libraries for grounded, knowledge-based responses.

Coordination & Autonomy

- **Pattern 6: Planning** [[18:19](#)] - Taking a complex goal and breaking it into a step-by-step dependency graph with specific milestones and resource constraints before execution.
- **Pattern 7: Multi-Agent Collaboration** [[20:49](#)] - Coordinating multiple specialized agents (e.g., a manager, researcher, and coder) to work together on shared tasks using common memory.
- **Pattern 14: Inter-Agent Communication** [[38:14](#)] - Establishing structured protocols, IDs, and expiration rules for how different AI models message and interact with one another securely.

Memory & Adaptation

- **Pattern 8: Memory Management** [[23:45](#)] - Classifying and indexing conversational history into short-term (episodic) and long-term storage to maintain contextual continuity over time.
- **Pattern 9: Learning & Adaptation** [[26:42](#)] - Collecting user feedback and outcome metrics to dynamically update prompts, policies, or system rules to prevent repeated mistakes.

Governance & Monitoring

- **Pattern 10: Goal Setting & Monitoring** [[29:17](#)] - Defining measurable key performance indicators (KPIs) and continuously tracking an agent's progress to identify and correct drift.

- **Pattern 11: Exception Handling & Recovery** [[31:34](#)] - Building safety nets to classify errors (temporary vs. permanent) and execute fallbacks or exponential back-offs when tools fail.
- **Pattern 12: Human-in-the-Loop** [[34:11](#)] - Implementing review queues for high-risk actions or edge cases, requiring human approval before the agent proceeds.
- **Pattern 17: Evaluation & Monitoring** [[49:57](#)] - Setting up golden test sets, SLAs, and continuous quality gates in production to detect performance anomalies over time.
- **Pattern 18: Guardrails & Safety** [[52:44](#)] - Sanitizing inputs, blocking injection attacks, and classifying risk levels to ensure ethical compliance, data security, and brand safety.

Advanced System Optimization

- **Pattern 15: Resource-Aware Optimization** [[43:08](#)] - Analyzing the complexity of a task and dynamically routing it to either a fast, cheap model or a powerful, expensive reasoning model based on cost budgets.
- **Pattern 16: Reasoning Techniques** [[46:35](#)] - Applying advanced logic paths (such as Chain of Thought, Tree of Thought, or Debate) to explore multiple solutions for highly complex, multi-variable problems.
- **Pattern 19: Prioritization** [[56:04](#)] - Building a dynamic dependency graph that scores and ranks tasks based on business value, risk, effort, and urgency, adjusting the queue on the fly.
- **Pattern 20: Exploration & Discovery** [[59:29](#)] - Mapping out broad knowledge spaces, clustering data into themes, and extracting specific insights, commonly used for deep research and R&D tasks.